

```
// 1.
Windsurf: Refactor | Explain | X
public void set(int index, int value) {
    ListNode current = front; // start at front of list

    for(int i = 0; i < index; i++) { // traverse through list until index
        current = current.next; // move to next node
    }

    current.data = value; // set data of node to value
}
```

```
// 2.
Windsurf: Refactor | Explain | X
public int lastIndex0f(int value) {
    ListNode current = front;
    int index = 0;
    int lastIndex = -1; // default value if value is not found

    // linked lists don't have built in indexes or known length, so a for loop can't be used
    // using a while loop instead allows us to naturally traverse through the list by following
    // the next references (pointers) until current is null, which indicates the end of the list
    while (current != null) { // traverse through list
        if (current.data == value) {
            lastIndex = index;
        }

        current = current.next; // move to next node
        index++; // move to next index
    }

    return lastIndex;
}
```

```

11
12     LinkedList list2 = new LinkedList(...elements: 1, 18, 2, 7, 18, 39, 18, 40);
13     int lastIndex = list2.lastIndexOf(value: 18);
14     System.out.println(lastIndex);
15 }
16
17

```

PROBLEMS 67 OUTPUT DEBUG CONSOLE TERMINAL TEST RESULTS ... Run: LinkedListMain

```

duey@MacBook-Air-Duy CS211 % /usr/bin/env /Library/Internet\ Plug-Ins/JavaAppletPlugin.plugin/Contents\
r/folders/w3/0x9k0j1s65gczs9x9ztngbw0000gn/T/cp_24312wgv5vmxwc5gf71z7ej5.jar week04.lab.LinkedList
6
duey@MacBook-Air-Duy CS211 %

```

3.

```
public boolean hasTwoConsecutive() {
    ListNode current = front;

    while(current != null && current.next != null) { // traverse through list and check if next
        node exists
        if(current.next.data == current.data + 1) { // if next node is 1 greater than current node
            return true; // found a consecutive pair so loop ends
        }

        current = current.next;
    }

    return false;
}
```

```
17      LinkedIntList list3 = new LinkedIntList(...elements: 1, 18, 2, 7, 8, 39, 18, 40);
18      System.out.println(list3.hasTwoConsecutive());
19
20      LinkedIntList list4 = new LinkedIntList(...elements: 1, 18, 17, 2, 7, 39, 18, 40, 8);
21      System.out.println(list4.hasTwoConsecutive());
22
23  }
```

PROBLEMS 67 OUTPUT DEBUG CONSOLE TERMINAL TEST RESULTS ... Run: LinkedIntMain

```
duey@MacBook-Air-Duy CS211 % /usr/bin/env /Library/Internet\ Plug-Ins/JavaAppletPlugin.plugin/Content
r/folders/w3/0x9k0j1s65gczs9x9ztngbw0000gn/T/cp_24312wgvg5vmxwc5gf71z7ej5.jar week04.lab.LinkedIntMa
true
false
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```

4.

```
public void switchPairs() {
    if (front == null || front.next == null) {
        return;
    }

    ListNode previous = null; // node before pair of nodes getting swapped
    ListNode current = front;

    front = current.next; // set front to next node

    // swap if there are at least 2 nodes left
    while (current != null && current.next != null) {
        ListNode next = current.next; // second node in pair
        ListNode after = next.next; // node after pair (start of next pair)

        // swap pairs
        next.next = current;
        current.next = after;

        // connect previous part of list to swapped pair
        if (previous != null) {
            previous.next = next;
        }

        // move to next pair
        previous = current; // current is now the second node in swapped pair
        current = after; // move to start of next pair
    }
}
```

5.

```
public void stutter() {
    ListNode current = front;

    while (current != null) {
        // current.next points to new duplicate nodes
        current.next = new ListNode(current.data, current.next);

        current = current.next.next; // move forward 2 nodes (move over the duplicate)
    }
}
```

```
26      LinkedIntList list5 = new LinkedIntList(...elements: 1, 8, 19, 4, 17);
27      list5.stutter();
28      System.out.println(list5);
29      System.out.println(list5.toStringShort());
```

PROBLEMS 67 OUTPUT DEBUG CONSOLE TERMINAL TEST RESULTS PORTS GITLENS Run

```
duey@MacBook-Air-Duy CS211 % /usr/bin/env /Library/Internet\ Plug-Ins/JavaAppletPlugin.plugin/Contents/Resources/Java/cp_24312wgv5vmxc5gf71z7ej5.jar week04.lab.L
front -> [1] -> [1] -> [8] -> [8] -> [19] -> [19] -> [4] -> [4] -> [17] -> [17] /
[1, 1, 8, 8, 19, 19, 4, 4, 17, 17]
```

6.

```
public void split() {
    ListNode negHead = null; // points to
    ListNode negTail = null; // points to
    ListNode posHead = null; // points to
    ListNode posTail = null; // points to

    ListNode current = front;

    while (current != null) {
        ListNode next = current.next; //
        current.next = null; // disconnect

        if (current.data < 0) {
            // add current node to negative list
            if (negHead == null) {
                negHead = current; // set
                negTail = current; // set
            }
            else { // since negHead isn't
                negTail.next = current; //
                negTail = current; // update
            }
        }
        else {
            // add current node to positive list
            if (posHead == null) {
                posHead = current; // set
                posTail = current; // set
            }
            else {
                posTail.next = current; //
                posTail = current; // update
            }
        }

        current = next; // move to next node
    }

    // connect the two lists
    if (negHead == null) {
        front = posHead; // if no negative list
    }
    else {
        front = negHead; // set front to negative list
        negTail.next = posHead; // connect
    }
}
```

```
31      LinkedIntList list6 = new LinkedIntList(...elements: 8, 7, -4, 19, 0, 43, -8, -7, 2);
32      list6.split();
33      System.out.println(list6);
34      System.out.println(list6.toStringShort());
```

PROBLEMS 67 OUTPUT DEBUG CONSOLE TERMINAL TEST RESULTS PORTS GITLENS Run: LinkedIntM

```
duey@MacBook-Air-Duy CS211 % /usr/bin/env /Library/Internet\ Plug-Ins/JavaAppletPlugin.plugin/Contents/Resources/Java/cp_24312wgv5vmxc5gf71z7ej5.jar week04.lab.L
front -> [-4] -> [-8] -> [-7] -> [8] -> [7] -> [19] -> [0] -> [43] -> [2] /
[-4, -8, -7, 8, 7, 19, 0, 43, 2]
```

7.

```
public void shift() {
    if (front == null || front.next == null) {
        return;
    }

    ListNode evenHead = front; // points to first even node
    ListNode oddHead = front.next; // points to first odd node

    ListNode even = evenHead; // tail of even list
    ListNode odd = oddHead; // tail of odd list

    while (odd != null && odd.next != null) {
        even.next = odd.next; // connect even to odd's next
        even = even.next; // move even tail forward
        odd.next = even.next; // connect odd to even's next
        odd = odd.next; // move odd tail forward
    }

    even.next = oddHead; // connect last even to first odd
}
```

```
36  LinkedIntList list7 = new LinkedIntList(...elements: 0, 1, 2, 3, 4, 5, 6, 7);
37  list7.shift();
38  System.out.println(list7);
39  System.out.println(list7.toStringShort());
40
41 }
```

PROBLEMS 67 OUTPUT DEBUG CONSOLE TERMINAL TEST RESULTS PORTS GITLENS

```
duey@MacBook-Air-Duy CS211 % /usr/bin/env /Library/Internet\ Plug-Ins/JavaAppletPlugin.plugin/Contents\ Resources/JavaPlugin2.jar week04.lab.LinkedIntMain
front -> [0] -> [2] -> [4] -> [6] -> [1] -> [3] -> [5] -> [7] /
[0, 2, 4, 6, 1, 3, 5, 7]
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```

8.

```
Windsurf: Refactor | Explain | X
public void reverse() {
    ListNode previous = null;
    ListNode current = front;

    while (current != null) {
        ListNode next = current.next;
        current.next = previous; // connect current to previous
        previous = current; // set previous to current
        current = next; // move to next
    }

    front = previous; // new front pointer
}
```

```
39  */
40
41  LinkedIntList list8 = new LinkedIntList(...elements: 1, 8, 19, 4, 17);
42  list8.reverse();
43  System.out.println(list8);
44  System.out.println(list8.toStringShort());
45
46 }
```

PROBLEMS 67 OUTPUT DEBUG CONSOLE TERMINAL TEST RESULTS PORTS GITLENS

```
duey@MacBook-Air-Duy CS211 % /usr/bin/env /Library/Internet\ Plug-Ins/JavaAppletPlugin.
w0000gn/T/cp_24312wgvg5vmxwc5gf71z7ej5.jar week04.lab.LinkedIntMain
front -> [17] -> [4] -> [19] -> [8] -> [1] /
[17, 4, 19, 8, 1]
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```